

Complexity and Certified Tree-Rearrangement Search for Structural-Entropy Hierarchies

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Abstract

Structural entropy scores a graph hierarchy by the description length induced by its encoding tree. Existing constructors return low-entropy trees, but two basic questions remain: which restricted optimization problem is already computationally hard, and how can a constructed tree be improved beyond local rotations without sacrificing exact scoring? We prove that balanced two-module structural-entropy minimization is NP-hard on simple unweighted cubic graphs through an exact affine equivalence with Minimum Bisection. We then derive exact entropy changes for rooted nearest-neighbor interchange (NNI) and subtree prune-and-regraft. The graft change has equivalent edge-LCA and changed-incidence forms; these identities show that NNI is a restricted graft and yield checkable certificates for the declared neighborhoods. They lead to NEST-G, which alternates exact NNI descent with exhaustive best-improvement grafting and fully rescores every accepted endpoint. Two independent graft evaluators agree with full entropy recomputation on 1,680 legal weighted-tree moves to within 1.5×10^{-15} bits. On five exact diagnostic instances where rotation-only search stopped above the optimum, grafting reaches the optimum in every case. Together, the complexity result, edit calculus, and certified search connect the global combinatorial obstacle to an auditable optimization procedure in encoding-tree space.

Keywords: structural entropy, hierarchical clustering, tree search, nearest-neighbor interchange, subtree grafting, graph bisection

1. Introduction

Many networks have meaningful organization at several resolutions. Vertices form local groups, related groups form larger modules, and the desired

output is therefore a hierarchy rather than a flat partition. Structural entropy expresses this organization as the description length of an encoding tree [1]. The objective has motivated hierarchy construction in graph clustering [2], chromatin-domain detection [3], and a growing range of graph-learning applications [4].

Optimizing the tree objective raises two different questions. The first is global: which restricted forms of structural-entropy minimization already contain a hard combinatorial problem? The second is geometric: once a tree has been constructed, which topology edits can expose an improvement and which certificate is obtained when search terminates? These questions are linked by the objective but require different tools. A complexity reduction identifies the hard partitioning core; an edit identity explains the local search space.

A preliminary conference manuscript addressed the second question with rooted nearest-neighbor interchange (NNI). Its method, NEST, derived an exact NNI entropy change, refined several initial trees, crossed selected two-move barriers, and compared the returned trees with exact optima on small graphs. NNI modifies one internal edge and is consequently inexpensive and easy to certify. Its locality is also a limitation: moving a subtree across several levels may require a sequence of rotations whose intermediate trees increase the objective and are rejected by monotone descent.

This article develops both questions beyond rotation-only search. For complexity, we isolate balanced two-module structural entropy and prove an exact equivalence with Minimum Bisection on regular graphs. For tree search, we add subtree prune-and-regraft, which relocates an entire subtree in one operation. The edge–LCA expansion gives an exact graft change, while a second derivation cancels every unchanged cluster–parent incidence. The two formulas support independent implementation checks and identify precisely where a graft can alter the objective.

Our contributions are as follows.

- We prove that the decision and optimization forms of balanced two-module structural entropy are NP-hard on simple unweighted cubic graphs through the identity $H^{\mathcal{P}}(G) = \log_2(n/2) + c(\mathcal{P})/|E|$.
- We give exact formulas for weighted NNI and rooted subtree grafting, including a path-supported cancellation theorem and the inclusion of the NNI neighborhood in the graft neighborhood.

- We introduce NEST-G, an exact best-improvement NNI-graft refiner. Exhaustive termination returns a checkable certificate for the declared graft neighborhood and, by inclusion, for the NNI neighborhood.
- We validate two independent graft-delta calculations against full recomputation on 1,680 moves and show, in a diagnostic exact audit, that graft search repairs all five residual failures of rotation-only NEST.
- We retain the preliminary study’s frozen 500-graph constructor study and 375-graph exact audit as the reference rotation-only evaluation, thereby separating inherited evidence from the new graft experiments.

The remainder of the article defines the problem family (Section 2), proves the constrained hardness result (Section 3), develops the edit calculus (Section 4), gives the certified algorithm (Section 5), and reports reproducible audits (Section 6).

2. Structural entropy and problem variants

Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. Write $d_u = \sum_v w_{uv}$, $M = \text{vol}(V) = \sum_u d_u$, and $W(A, B) = \sum_{u \in A, v \in B} w_{uv}$ for disjoint vertex sets. For $S \subseteq V$, let $V_S = \sum_{u \in S} d_u$ and let g_S be the weight of edges with exactly one endpoint in S .

An encoding tree T has one leaf for every vertex. Each tree node a represents the cluster S_a of descendant leaves, and a^- denotes its parent. The structural entropy of G under T is

$$H^T(G) = - \sum_{a \neq r} \frac{g_{S_a}}{M} \log_2 \frac{V_{S_a}}{V_{S_a^-}}. \quad (1)$$

Lower values indicate shorter graph-dependent descriptions. Zero-volume components contribute zero and may be attached after optimizing the positive-volume part, so the theory below assumes positive volume whenever a logarithm is evaluated.

Two equivalent views will be useful. The node view is Eq. (1). In the edge view, every edge is charged at the least common ancestor of its endpoints:

$$H^T(G) = \sum_{uv \in E} \frac{w_{uv}}{M} \log_2 \frac{V_{\text{lca}_T(u,v)}^2}{d_u d_v}. \quad (2)$$

The equivalence follows by expanding each cut term and telescoping the code lengths along the two leaf-to-LCA paths.

We distinguish three optimization problems.

Problem 1 (Tree structural entropy). Given G , find an encoding tree minimizing Eq. (1).

Problem 2 (Two-dimensional structural entropy). Given G , find a height-two encoding tree, equivalently a nonempty partition $\mathcal{P} = \{S_1, \dots, S_K\}$, minimizing structural entropy.

Problem 3 (Balanced two-module structural entropy). Given a regular graph of even order, minimize two-dimensional structural entropy over partitions $\{S, V \setminus S\}$ satisfying $|S| = |V|/2$.

The third problem is a restriction of the second. We state it separately because its balance constraint is essential to the reduction in Section 3.

For search, we work with rooted binary trees. Prior work establishes that every encoding tree admits a binary refinement of no greater structural entropy [3, 2]. Thus a global optimum exists in the binary class, although constructors may initially return multiway trees and refine them before search.

3. A hard balanced partitioning core

This section isolates a constrained form of structural entropy that is already computationally hard. The result also explains why cut quality alone does not characterize a general two-dimensional optimum: module volumes enter the objective unless they are controlled.

We use the following decision form. An instance consists of a simple cubic graph $G = (V, E)$ with even $n = |V|$ and an integer b . The question is whether there exists $S \subset V$ with $|S| = n/2$ such that the height-two tree induced by $\{S, V \setminus S\}$ has entropy at most $\log_2(n/2) + b/|E|$. The threshold is represented by the integer pair (n, b) ; no numerical approximation to the logarithm is part of the input.

For a partition $\mathcal{P} = \{S_1, \dots, S_K\}$, let e_i be the internal edge weight of S_i . Directly expanding Eq. (1) gives

$$\begin{aligned} H^{\mathcal{P}}(G) &= -\frac{1}{M} \sum_{u \in V} d_u \log_2 \frac{d_u}{V_{S(u)}} - \frac{1}{M} \sum_{i=1}^K g_i \log_2 \frac{V_i}{M} \\ &= C_G + \frac{1}{M} \sum_{i=1}^K [2e_i \log_2 V_i + g_i \log_2 M], \end{aligned} \quad (3)$$

where $C_G = -M^{-1} \sum_u d_u \log_2 d_u$ does not depend on the partition and $V_i - g_i = 2e_i$.

Lemma 4 (Balanced regular identity). *Let G be a loop-free d -regular graph with even order n and $m = |E| = dn/2$. For a balanced bipartition $\mathcal{P}$ with cut size $c(\mathcal{P})$,*

$$H^{\mathcal{P}}(G) = \log_2 \frac{n}{2} + \frac{c(\mathcal{P})}{m}. \quad (4)$$

Consequently, for two balanced bipartitions $\mathcal{P}$ and $\mathcal{Q}$, $H^{\mathcal{P}}(G) - H^{\mathcal{Q}}(G) = [c(\mathcal{P}) - c(\mathcal{Q})]/m$.

Proof. Every vertex has degree d , both modules have volume $dn/2 = M/2$, and both module cuts equal $c(\mathcal{P})$. The leaf contribution to the height-two tree is

$$-\frac{1}{M} \sum_{u \in V} d \log_2 \frac{d}{M/2} = \log_2(n/2).$$

The two module contributions sum to $-2c(\mathcal{P})M^{-1} \log_2(1/2) = c(\mathcal{P})/m$. Adding the terms proves Eq. (4). $\square$

Theorem 5 (Constrained structural-entropy hardness). *The decision form of balanced two-module structural entropy is NP-complete on simple unweighted cubic graphs. Consequently, computing an optimal balanced two-module encoding tree is NP-hard on the same class.*

Proof. Minimum Bisection is NP-complete on simple cubic graphs [5]. Map an instance (G, b) identically to the entropy decision instance (G, b) . The mapping is polynomial and introduces neither weights nor auxiliary vertices. By Lemma 4, a balanced partition $\mathcal{P}$ satisfies $H^{\mathcal{P}}(G) \leq \log_2(n/2) + b/m$ if and only if $c(\mathcal{P}) \leq b$. This proves NP-hardness. A balanced set S is a polynomial certificate, and the equivalent integer comparison $c(\mathcal{P}) \leq b$ verifies

it in polynomial time; hence the decision problem is in NP. An optimizer would decide the problem by comparing its returned cut size with b , so the optimization form is NP-hard. $\square$

The balance constraint cannot simply be discarded. On the 10-vertex pentagonal prism and Möbius ladder, exhaustive enumeration finds a 4/6 partition with entropy 2.625497 bits, below the best balanced value 2.655261. Thus regularity and exactly two nonempty modules do not force a bisection. This counterexample delineates the theorem’s scope and rules out an otherwise tempting transfer to unconstrained two-module optimization.

4. Exact calculus for local and nonlocal tree edits

4.1. Rooted nearest-neighbor interchange

Consider three disjoint subtrees A, B, C inside a local cluster $P = A \cup B \cup C$. A rooted NNI replaces $((A, B), C)$ by $(A, (B, C))$. Let $X = A \cup B$ and $Y = B \cup C$.

Proposition 6 (Exact weighted NNI delta). *For $V_X, V_Y, V_P > 0$, the entropy change is*

$$\Delta_{\text{NNI}} H^T = \frac{2}{M} \left[W(A, B) \log_2 \frac{V_P}{V_X} + W(B, C) \log_2 \frac{V_Y}{V_P} \right]. \quad (5)$$

The other rooted alternative follows by exchanging A and B .

Proof. Only the code lengths involving A, B, C, X , and Y change. Substitute $g_X = g_A + g_B - 2W(A, B)$ and $g_Y = g_B + g_C - 2W(B, C)$ into Eq. (1). The coefficients of g_A, g_B, g_C telescope, leaving Eq. (5). $\square$

The identity permits exhaustive best-improvement descent: enumerate every eligible internal edge, commit the most negative exact delta, and stop when no delta is negative. Termination certifies one-NNI local optimality for the returned tree.

4.2. Rooted subtree prune-and-regraft

A legal graft chooses a non-root source subtree S and a target subtree Q that is neither an ancestor nor a descendant of S . It detaches S , suppresses the resulting unary parent, and replaces Q by a new binary parent with children S and Q . Same-parent choices are omitted because they recreate the same unordered hierarchy. Figure 1 contrasts the short-range and long-range edits.

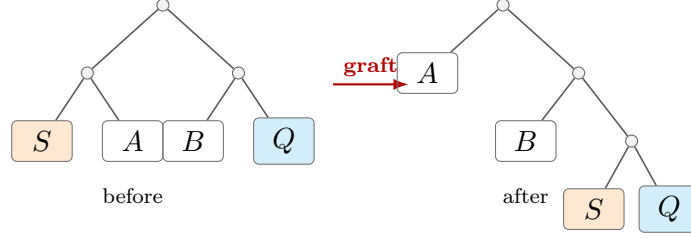


Figure 1: A graft detaches the source subtree S and inserts it next to target Q , suppressing the old unary parent. Unlike one NNI, the source and target may be separated by several tree levels.

Theorem 7 (Exact edge-LCA graft delta). *Let T' be obtained from T by a legal graft. Then*

$$H^{T'}(G) - H^T(G) = \frac{2}{M} \sum_{uv \in E} w_{uv} \log_2 \frac{V_{\text{lca}_{T'}(u,v)}}{V_{\text{lca}_T(u,v)}}. \quad (6)$$

Every edge whose LCA cluster and LCA volume are unchanged contributes zero.

Proof. Apply Eq. (2) to T' and T . The factors $d_u d_v$ depend only on the graph and cancel. Combining the two logarithms yields Eq. (6). $\square$

The all-edge sum is a correctness-oriented formula. A second exact form makes the locality explicit. Let $\mathcal{C}(T)$ be the non-root clusters of T and $p_T(C)$ the parent cluster of C . Define

$$\psi(C, P) = -\frac{g_C}{M} \log_2 \frac{V_C}{V_P}.$$

Let $I(T) = \{(C, p_T(C)) : C \in \mathcal{C}(T)\}$ be the set of non-root cluster-parent incidences. For a graft from T to T' , define $\mathcal{A}_- = I(T) \setminus I(T')$ and $\mathcal{A}_+ = I(T') \setminus I(T)$.

Theorem 8 (Changed-incidence graft formula). *For a legal graft from T to T' ,*

$$H^{T'}(G) - H^T(G) = \sum_{(C,P) \in \mathcal{A}_+} \psi(C, P) - \sum_{(C,P) \in \mathcal{A}_-} \psi(C, P). \quad (7)$$

Every incidence in $\mathcal{A}_- \cup \mathcal{A}_+$ lies on the source-removal path, the target-insertion path, or at the new parent of S and Q .

Proof. Write Eq. (1) as the sum of $\psi(C, p_T(C))$ over $C \in \mathcal{C}(T)$. Detaching S changes only ancestors between the old parent of S and the first common ancestor of the source and target paths; inserting S changes only the corresponding target ancestors and creates the parent $S \cup Q$. Every cluster and parent outside these paths has the same leaf set before and after the edit, so its incidence belongs to $I(T) \cap I(T')$ and cancels. The noncancelling old and new terms are exactly $\mathcal{A}_-$ and $\mathcal{A}_+$, which proves both the identity and the path support statement. $\square$

The theorem is stronger than a cut-edge-only shortcut. When S is inserted, ancestor clusters on the target path gain its volume; when S is removed, clusters on the source path lose it. Therefore an edge with neither endpoint in S can still change contribution when its LCA lies on an affected path.

4.3. Relation between the neighborhoods

Proposition 9 (NNI is a restricted graft). *Every rooted NNI neighbor of a binary tree is also a legal rooted graft neighbor.*

Proof. For $((A, B), C) \rightarrow (A, (B, C))$, detach C and graft it next to B . Suppressing the old parent exposes (A, B) , and inserting the new parent of B, C produces $(A, (B, C))$. Grafting C next to A gives the symmetric NNI alternative. $\square$

Corollary 10. *A tree with no improving legal graft has no improving rooted NNI.*

The converse fails on the audited instances in Section 6.3: trees certified against one-step NNI admit an improving graft. Grafting thus tests a strictly richer neighborhood on those instances rather than merely rescoring the same rotations.

Table 1: Relation between the two exact edit neighborhoods. Tree distance is the maximum source relocation accomplished by one move. A certificate means that all legal candidates in the stated neighborhood have nonnegative exact entropy change.

Edit	Candidates	Range	Exhaustive stopping certificate
Rooted NNI	$O(n)$	one edge	one-NNI local optimum
Rooted graft	$O(n^2)$	any legal path	graft-local optimum (and therefore one-NNI local)

Algorithm 1: Exact certified NNI-graft refinement

Input: Graph G , binary tree T , tolerance ε

- 1 fully rescore T and set $h \leftarrow H^T(G)$;
- 2 **repeat**
- 3 repeatedly commit the most negative exact NNI delta until none is below $-\varepsilon$; fully verify every committed tree;
- 4 enumerate every legal ordered source–target graft of the current tree;
- 5 evaluate each candidate by Eq. (6) or Eq. (7);
- 6 **if** *the best fully rescored graft endpoint is below $h - \varepsilon$* **then**
- 7 commit it and update h ;
- 8 **else**
- 9 **break**;
- 10 **until** *no accepted graft exists*;
- 11 **return** T , h , and the final neighborhood audit

5. Certified NNI-graft search

NEST-G separates tree construction from topology refinement. It accepts any rooted binary initializer, performs exact NNI descent, searches the complete legal graft neighborhood, and reruns NNI descent after each accepted graft. Every committed endpoint is independently rescored with Eq. (1).

Proposition 11 (Monotonicity and termination). *Algorithm 1 monotonically decreases structural entropy and terminates after finitely many accepted moves.*

Proof. The algorithm commits a move only after its independently recomputed endpoint is at least ε below the current objective. The number of rooted binary trees on a fixed labeled leaf set is finite, and strict decrease prevents revisiting a tree. $\square$

Proposition 12 (Declared-neighborhood certificate). *If Algorithm 1 reaches its natural stopping condition, its output has neither an improving rooted NNI nor an improving legal graft under the stated tolerance.*

Proof. The final NNI descent exhausts all rooted NNI candidates. The subsequent graft sweep exhausts all legal ordered source–target pairs and finds none whose verified endpoint improves the objective. Proposition 9 also

implies the NNI part of the certificate from exhaustive graft locality, but we retain the explicit NNI sweep because its closed-form candidates are cheaper to evaluate. $\square$

For n leaves there are $O(n)$ rooted NNI candidates and $O(n^2)$ ordered graft candidates per sweep. A correctness-first graft candidate is rebuilt and rescored in $O(n + |E|)$ time, giving $O(n^2(n + |E|))$ time for one exhaustive reference sweep. The independent changed-incidence evaluator instead recomputes only terms whose cluster-parent relation changes. Cached volumes, cuts, and LCA buckets can replace both reference scans without changing the identities. Candidate grafts are independent at a fixed tree, so a sweep can also be evaluated in parallel. The present implementation uses the slower reference path so that the certificate does not depend on an unvalidated cache.

For small graphs, the final endpoint is compared with an independent exact subset dynamic program. For every nonempty $S \subseteq V$, let $F(S)$ be the minimum contribution below a binary node representing S . Then

$$F(\{v\}) = 0, \quad F(S) = \min_{A \dot{\cup} B = S} \left\{ F(A) + F(B) - \frac{g_A}{M} \log_2 \frac{V_A}{V_S} - \frac{g_B}{M} \log_2 \frac{V_B}{V_S} \right\}. \quad (8)$$

Enumerating unordered splits gives $O(3^n)$ time and $O(2^n)$ memory. Binary sufficiency makes $F(V)$ a global structural-entropy optimum [3, 2].

6. Computational verification and empirical audits

The evaluation is organized by claim dependency. We first check the algebraic identity used in the reduction, then validate the new graft operator, next examine the five known rotation-only misses, and finally summarize the frozen broad and exact benchmarks inherited from the preliminary study. All selections use structural entropy only; exact optima are consulted after selection.

6.1. Finite checks of the balanced identity

An independent implementation enumerated 3,199 unordered balanced partitions across 23 regular graphs drawn from cycles, complete graphs, complete bipartite graphs, prisms, Möbius ladders, and circulant families. The direct two-dimensional entropy and Eq. (4) agreed within 1.34×10^{-15} bits. Entropy minimizers and minimum bisections agreed on every graph.

A separate enumeration of 13,697 unrestricted bipartitions found the two 10-vertex counterexamples reported in Section 3. The finite check supports the algebra and guards against an invalid removal of the balance constraint; Theorem 5 is established by the proof rather than by enumeration.

6.2. Independent validation of graft deltas

We generate 20 connected weighted graphs with five to nine vertices and one random binary tree per graph. For every legal graft, we compare three values: full recomputation of Eq. (1), the edge-LCA delta in Eq. (6), and the changed-incidence delta in Eq. (7). Across 1,680 legal moves, both delta evaluators match full recomputation to floating-point precision; the largest absolute discrepancy is below 1.5×10^{-15} bits. Every candidate preserves a rooted binary topology and exactly one copy of every leaf.

Four additional best-improvement runs check end-to-end behavior. Each accepted graft strictly reduces the independently rescored objective, and exhaustive enumeration at termination finds no improving legal graft among 88–124 final candidates. These tests validate the reference implementation and the stated neighborhood certificate separately from the method’s optimization quality.

6.3. Repairing the residual exact misses

The earlier sealed exact audit contained five instances on which 32 coalescent starts followed by NNI and bounded two-step search missed the exact dynamic-programming optimum. We reconstruct the same graphs and restart streams, apply exhaustive best-improvement grafting to each returned tree, and consult the exact optimum only after selecting the lowest-entropy endpoint. Table 2 reports the diagnostic result.

Grafting reaches the exact optimum on all five cases. It improves between 18 and 31 of the 32 starts per graph, showing that the outcome is not confined to one fortuitous restart. These instances were selected precisely because rotation-only search missed them; the result therefore demonstrates the mechanism and a strict neighborhood advantage on the diagnosed cases. It is not used to estimate the frequency of improvement on unseen graphs.

6.4. Frozen rotation-only benchmark

The preliminary study contains 100 graphs from each of five 64-vertex hierarchical stochastic block-model regimes: clean, noisy, imbalanced, weighted, and weak hierarchy. NEST uses a fixed pool of SE agglomeration, recursive

Table 2: Diagnostic graft audit on all five known residual misses. “Improved starts” counts the original 32 rotation-refined starts whose entropy is further reduced by grafting. The final gap is measured against the independent subset-DP optimum; lower is better.

n	Regime	Seed	NNI gap (bits)	Improved starts	Graft gap
12	Clean	168	0.001323	31/32	0
14	Weak hierarchy	3004	0.002440	18/32	0
16	Noisy	4001	0.005274	23/32	0
16	Noisy	4002	0.007208	28/32	0
16	Weak hierarchy	4002	0.010305	26/32	0

SE, and Paris initial trees, refines each by exact NNI and bounded two-move search, and returns the lowest independently rescored endpoint. HCSE and BBM are external comparators and are not added to the initializer pool. Table 3 reproduces the frozen results.

Table 3: Mean tree structural entropy H^T (bits) on 100 paired graphs per HSBM regime. The $\pm$ value is a 95% t -interval over graph seeds; lower is better, and bold marks the lowest column mean. $\text{BBM}^\dagger$ receives the planted fine-cluster count; the other methods do not use labels.

Method	Clean	Noisy	Imbalanced	Weighted	Weak hierarchy
SE-agglomerative	2.899 ± 0.021	3.723 ± 0.017	3.162 ± 0.023	2.445 ± 0.016	3.461 ± 0.022
Louvain-2L	4.109 ± 0.022	4.603 ± 0.013	4.241 ± 0.019	3.657 ± 0.032	4.432 ± 0.018
Paris	2.869 ± 0.022	3.709 ± 0.017	3.129 ± 0.023	2.349 ± 0.015	3.438 ± 0.023
HCSE	3.228 ± 0.031	4.485 ± 0.068	3.598 ± 0.036	2.741 ± 0.029	4.032 ± 0.051
$\text{BBM}^\dagger$	3.335 ± 0.021	4.118 ± 0.020	3.573 ± 0.025	3.007 ± 0.020	3.924 ± 0.022
NEST (ours)	2.827 ± 0.021	3.683 ± 0.017	3.086 ± 0.023	2.340 ± 0.015	3.414 ± 0.021

Across the 500 paired graphs, rotation-only NEST lowers entropy by 0.4239 ± 0.0086 bits relative to the better of HCSE and BBM and wins every paired comparison. Figure 2 shows the per-instance margins.

6.5. Frozen exact-optimum audit

After disjoint calibration fixed 32 candidates per method, the exact audit sealed 250 graphs at $n = 12$, 100 at $n = 14$, and 25 at $n = 16$. The exact subset dynamic program is used only for evaluation. Table 4 reports the rotation-only results that produced the five diagnostic cases above.

NEST improves over the stronger external SE baseline on 500/500 graphs

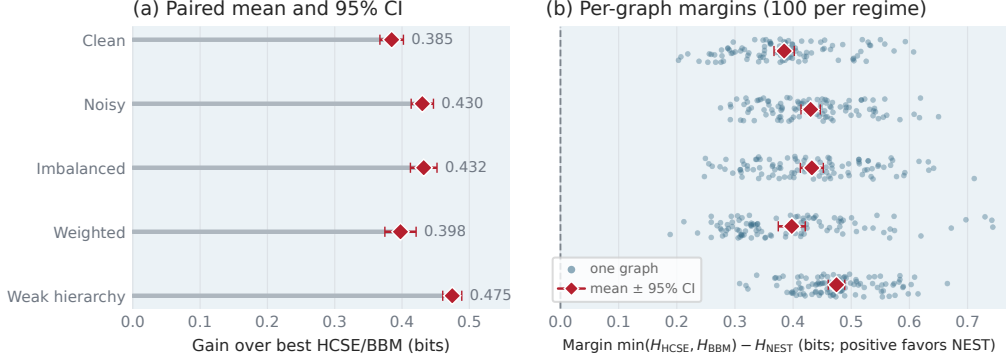


Figure 2: Rotation-only NEST against the stronger of HCSE and BBM on each graph. A positive margin means that NEST has lower structural entropy. The left panel reports regime means and paired 95% confidence intervals. In the right panel, each blue point is one graph, red diamonds are regime means, and horizontal whiskers are their paired 95% confidence intervals.

Table 4: Sealed exact-optimum audit on independently generated HSBMs with n vertices and 32 candidates per method: coalescent starts for NEST, target heights for HCSE, and label-free cluster counts for BBM. Exact columns report optimum hits over the listed graphs. The NEST interval is an exact 95% Clopper–Pearson interval for its hit rate. Its relative gap is $100(H_{\text{NEST}}^T - H^*)/H^*$, reported as mean/worst; zero is optimal. Candidate selection does not use H^* .

n	Graphs	Exact optima found			NEST diagnostics	
		NEST	HCSE	BBM	95% hit-rate interval	Mean / worst relative gap
12	250	249/250	2/250	1/250	[97.79, 99.99]%	0.00032% / 0.07990%
14	100	99/100	0/100	0/100	[94.55, 99.97]%	0.00121% / 0.12060%
16	25	22/25	0/25	0/25	[68.78, 97.45]%	0.03990% / 0.44934%

Rotation-only NEST reaches 370 of 375 exact optima, compared with 2/375 for HCSE and 1/375 for label-free BBM under the declared 32-candidate protocol. The graft audit closes these five observed NEST gaps. A disjoint confirmatory protocol has been frozen before execution on graph seeds 500–519 in each of the five regimes. It keeps the 32-start rotation-only endpoint as a paired control, adds exhaustive graft refinement without label-based

selection, and computes the exact optimum only after both endpoints have been selected. Its primary endpoints are the paired exact-hit difference and the paired entropy change; runtime and candidate counts are secondary endpoints. Results from that protocol will replace this paragraph only after the frozen artifact passes integrity checks.

7. Related work

Structural entropy. Structural information was introduced to measure network organization with an encoding tree [1]. HCSE and BBM construct structural-entropy hierarchies through partitioning, merging, and balancing operations [2]. SuperTAD establishes binary sufficiency and develops contiguity-aware dynamic programs for chromatin hierarchies [3]. Our work instead studies the complexity of a balanced partitioning restriction and exact topology edits in arbitrary weighted graph hierarchies. The broader theory and applications are reviewed in [4].

Graph hierarchical clustering. Paris constructs graph dendrograms through node-pair sampling [6], while Louvain optimizes a flat modularity objective [7]. Dasgupta’s hierarchical cost and the Moseley–Wang revenue objective have motivated algorithms and approximation analyses [8, 9]. Their objectives differ from structural entropy, but they share the principle that an edge contribution depends on a cluster associated with its endpoints. Section 8 identifies edge–LCA objectives as the natural domain for extending our edit calculus.

Tree rearrangement. NNI is a classical local move in phylogenetic tree space, where even the distance between two trees is nontrivial [10]. Local NNI search has also been studied for hierarchical-clustering costs [11]. PERCH uses rotations to preserve purity during online construction [12]. GRINCH adds grafting to repair mistakes that cannot be corrected by local rotations alone [13]. Our contribution is an objective-specific exact calculus: both NNI and graft are scored against weighted graph structural entropy, every accepted endpoint is independently verified, and exhaustive termination certifies the declared neighborhood.

Exact hierarchy optimization and graph partitioning. Cluster trellises organize exact inference over subset states [14]; our dynamic program specializes this principle to structural entropy. The hardness argument draws on the

established complexity of graph bisection in regular graphs [5] and uses an exact affine identity rather than an approximation between objectives.

8. Discussion and extensions

The results expose a useful division of labor. The balanced-bisection theorem shows that a hard cut problem is already present in a restricted two-dimensional objective. The edit theorems then provide exact, auditable progress within the much larger encoding-tree space. Hardness motivates certified approximation by search; local certificates state precisely what that search has exhausted.

Grafting enlarges the neighborhood without changing the objective. One NNI is a particular graft, while a long-range graft can relocate a subtree across several levels in one committed step. On the five diagnosed cases, rotation-only search had reached a one-NNI certificate while the larger graft neighborhood still contained decreasing moves. The path formula further indicates how to scale the operation. Cached module volumes, cuts, and affected LCA buckets can replace full candidate reconstruction, and candidates at a fixed tree can be scored in parallel on CPU or GPU.

The edge-LCA derivation also suggests a broader objective class. Any objective of the form

$$F_\phi(T; G) = \sum_{uv \in E} w_{uv} \phi(V_{\text{lca}_T(u,v)}, d_u, d_v; G)$$

admits a graft difference obtained by subtracting the old and new LCA terms. Whether that difference has a compact path cache depends on the sufficient statistics used by ϕ . Dasgupta cost and Moseley–Wang revenue are natural test cases, but a general theorem should be claimed only after the formulas and implementations are verified for non-equivalent objectives.

The present evidence has two boundaries. The hardness theorem concerns a balanced height-two restriction, whereas the search algorithm operates over unrestricted binary encoding trees. The exact formulas connect these parts through one objective but do not turn the constrained hardness result into a hardness theorem for unrestricted tree optimization. In addition, the five graft repairs are diagnostic cases; the frozen disjoint suite is needed to estimate how often the larger neighborhood helps beyond those cases.

Three theoretical questions remain especially important. First, transferring Theorem 5 to unconstrained two-dimensional or unrestricted tree structural

entropy requires a valid balance-forcing or normal-form construction. Second, the number and barrier profile of NNIs needed to simulate a graft should be bounded in terms of source–target tree distance. Third, restricted graph families may admit polynomial algorithms or stronger optimality certificates. These questions define the next stage of the journal analysis rather than changing the proved claims above.

9. Conclusion

We connected a hard partitioning core with exact optimization geometry for structural-entropy hierarchies. Balanced two-module optimization is NP-hard on cubic graphs through an exact equivalence with Minimum Bisection. Within tree space, exact NNI and graft identities support monotone search and explicit neighborhood certificates. The graft neighborhood contains every NNI move and repairs all five diagnosed failures of rotation-only search. Independent formulas, full recomputation, exact dynamic programming, and frozen manifests make each layer of evidence separately checkable. The remaining empirical question is quantified by a disjoint, frozen graft-confirmation protocol; algorithmic scaling requires replacing the correctness-first full rescoring with validated path caches and parallel candidate evaluation.

Data and code availability

The implementation, deterministic validators, frozen manifests, and raw per-instance records are maintained in the project repository. Each numerical claim in this article is generated from a versioned JSON artifact and can be recomputed by the accompanying verification scripts. A permanent archival identifier will be added to the accepted version.

Declaration of generative AI and AI-assisted technologies in the writing process

During manuscript preparation, the authors used OpenAI Codex to assist with manuscript organization, language editing, code review, and mechanical consistency checks. The authors reviewed and edited all generated material, independently checked the proofs, references, code, and numerical results, and take full responsibility for the content of the article.

References

- [1] A. Li, Y. Pan, Structural information and dynamical complexity of networks, *IEEE Transactions on Information Theory* 62 (6) (2016) 3290–3339.
- [2] Y. Pan, B. Fan, P. Long, F. Zheng, An information-theoretic perspective of hierarchical clustering on graphs, in: *Proceedings of UAI*, Vol. 286, PMLR, 2025, pp. 3322–3345.
- [3] Y. W. Zhang, M. B. Wang, S. C. Li, SuperTAD: Robust detection of hierarchical topologically associated domains with optimized structural information, *Genome Biology* 22 (1) (2021) 45. doi:10.1186/s13059-020-02234-6.
- [4] D. Su, H. Peng, Y. Pan, A. Li, A survey of structural entropy: Theory, methods, and applications, in: *Proceedings of IJCAI*, 2025, pp. 10660–10668.
- [5] T. N. Bui, S. Chaudhuri, F. T. Leighton, M. Sipser, Graph bisection algorithms with good average case behavior, *Combinatorica* 7 (2) (1987) 171–191. doi:10.1007/BF02579448.
- [6] T. Bonald, B. Charpentier, A. Galland, A. Hollocou, Hierarchical graph clustering using node pair sampling, in: *Advances in Neural Information Processing Systems*, 2018, pp. 8214–8223.
- [7] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, E. Lefebvre, Fast unfolding of communities in large networks, *Journal of Statistical Mechanics: Theory and Experiment* 2008 (10) (2008) P10008. doi:10.1088/1742-5468/2008/10/P10008.
- [8] S. Dasgupta, A cost function for similarity-based hierarchical clustering, in: *Proceedings of STOC*, 2016, pp. 118–127. doi:10.1145/2897518.2897527.
- [9] B. Moseley, J. R. Wang, Approximation bounds for hierarchical clustering: Average linkage, bisecting k-means, and local search, in: *Advances in Neural Information Processing Systems*, Vol. 30, 2017, pp. 3094–3103.

- [10] M. Li, J. Tromp, L. Zhang, On the nearest neighbour interchange distance between evolutionary trees, *Journal of Theoretical Biology* 182 (4) (1996) 463–467.
- [11] H. Jowhari, Hierarchical clustering via local search, *arXiv preprint arXiv:2405.15983* (2024).
- [12] A. Kobren, N. Monath, A. Krishnamurthy, A. McCallum, A hierarchical algorithm for extreme clustering, in: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 255–264. doi:10.1145/3097983.3098079.
- [13] N. Monath, A. Kobren, A. Krishnamurthy, M. R. Glass, A. McCallum, Scalable hierarchical clustering with tree grafting, in: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 1438–1448. doi:10.1145/3292500.3330929.
- [14] S. Macaluso, C. Greenberg, N. Monath, J. A. Lee, P. Flaherty, K. Cranmer, A. McGregor, A. McCallum, Cluster trellis: Data structures and algorithms for exact inference in hierarchical clustering, in: *Proceedings of AISTATS*, Vol. 130, PMLR, 2021, pp. 2467–2475.

Appendix A. Additional derivations and audit details

Appendix A.1. Derivation of the partition form

For a height-two partition, split Eq. (1) into leaf and module terms:

$$H^{\mathcal{P}}(G) = -\frac{1}{M} \sum_u d_u \log_2 \frac{d_u}{V_{S(u)}} - \frac{1}{M} \sum_i g_i \log_2 \frac{V_i}{M}.$$

Expanding the logarithms and collecting the graph-only term gives

$$C_G + \frac{1}{M} \sum_i [(V_i - g_i) \log_2 V_i + g_i \log_2 M].$$

Because $V_i = 2e_i + g_i$ in an undirected graph, this is Eq. (3).

Appendix A.2. Derivation of the edge-LCA expansion

For one edge uv , its weight appears in the cut g_{S_a} precisely when S_a contains one endpoint but not the other. Along the u side, the corresponding logarithms telescope from d_u to $V_{\text{lca}_T(u,v)}$; the v side telescopes analogously. Hence the contribution of uv is

$$\frac{w_{uv}}{M} \left[\log_2 \frac{V_{\text{lca}_T(u,v)}}{d_u} + \log_2 \frac{V_{\text{lca}_T(u,v)}}{d_v} \right],$$

which equals its term in Eq. (2).

Appendix A.3. Why a source-cut shortcut is insufficient

It is tempting to evaluate a graft using only edges crossing the relocated subtree S . This is incorrect. An ancestor on the insertion path gains the volume of S , while an ancestor on the removal path loses it. An edge with both endpoints outside S may have one of these ancestors as its old or new LCA, and its logarithmic contribution can therefore change. The deterministic validator contains a counterexample in which the cut-only shortcut predicts -0.119611 bits while full recomputation gives $+0.106521$ bits. The edge-LCA and changed-incidence formulas both return the latter value.

Appendix A.4. Exact-audit protocol for the five repaired cases

The five graphs are identified by the frozen $n = 12, 14, 16$ manifests. For each graph, the diagnostic script reconstructs the original 32 pairwise-coalescent starts, runs the same NNI and bounded two-step refinement, and verifies the recorded pre-graft minimum. It then applies exhaustive best-improvement graft refinement with NNI descent after every accepted graft. Candidate selection is based only on structural entropy. The exact dynamic-programming value is queried after selecting the best of the 32 endpoints. The per-restart traces, accepted graft counts, elapsed times, and before/after objectives are stored in the accompanying JSON artifact.